Predicting NBA Playoff Success

from Regular-Season Data

A machine-learning rebuild of our Fall 2025 Excel model, and a measure of how much regular-season data can predict.

PRESENTED BY

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29	862	0.72
seasons	team-seasons	cross-validated R^2

RECAP

The Story So Far

In Fall 2025, we built an Excel-based playoff predictor for MBA 503.

FINDING 1

3P% Wins Among Basic Stats

Among traditional box-score stats, three-point percentage was the strongest positive predictor of playoff success.

FINDING 2

TS% Beats 3P%

True Shooting absorbed three-point percentage by capturing the full scoring profile — twos, threes, and free throws.

FINDING 3

Net Rating Reigns

Net Rating dominated everything. A one-SD increase predicted +0.8 playoff rounds. Final R^2 = 0.52.

*The conclusion was clean: **substance beats style.** But the model only explained 52% of variance.*

THE OPEN QUESTION

Was 0.52 the floor or the ceiling?

If we used 29 seasons instead of 11, Python instead of Excel, and machine learning instead of OLS,

Could we explain the unexplained 48%?

SPOILER

Partly. Cross-validated R^2 rose to 0.72, but about 28% stayed out of reach.

PRE-REGISTERED PREDICTIONS

Three Hypotheses

H1

Win Percentage is Sufficient

Win % alone will capture the majority of explainable variance. Everything else competes for the leftovers.

H2

Advanced Stats Add Marginally

Net Rating, TS%, and pace adjustments improve R^2 modestly. They explain **how** teams win, not whether they win.

H3

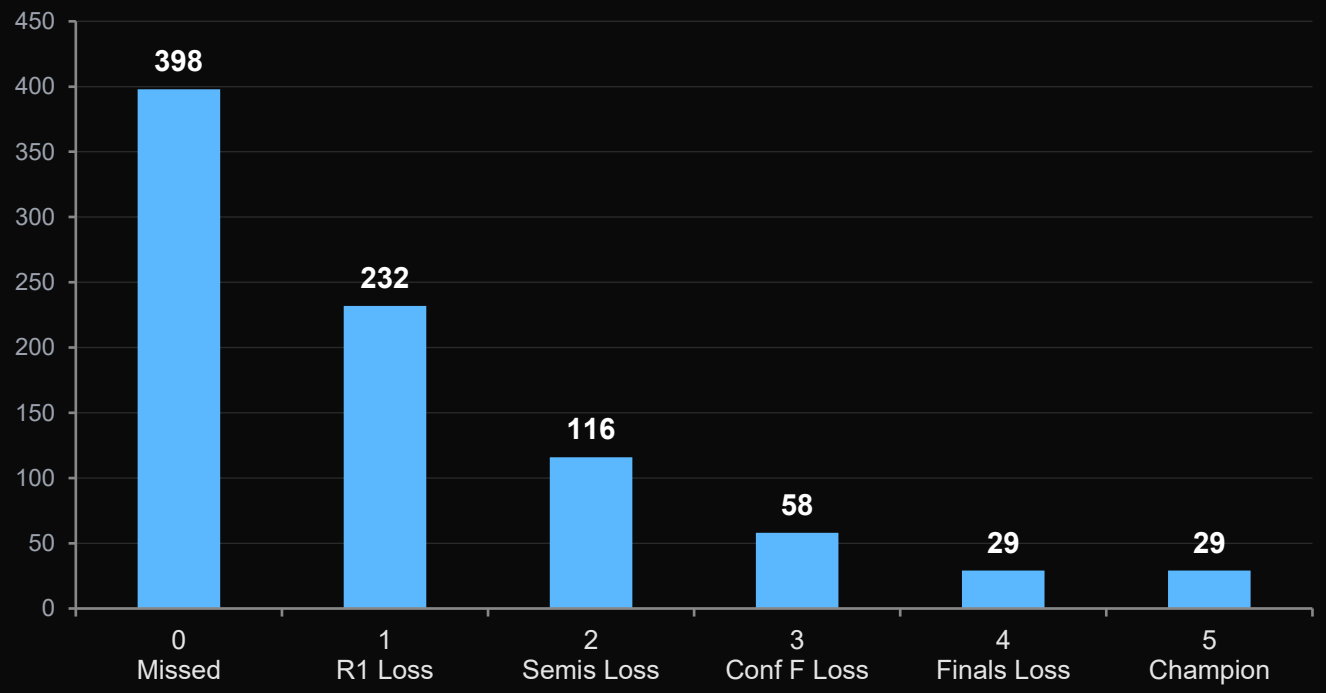
Machine Learning Beats Linear Regression

XGBoost will outperform linear regression by capturing interactions.

862 Team-Seasons, Heavily Skewed

Most teams don't make deep playoff runs. That's what makes prediction hard.

Playoff Rounds Distribution (1997-2025)



17 Features in 3 Buckets

Team Performance (14)

Win%, Net Rating, TS%, eFG%, ORtg, DRtg, 3P%, FT%, ORB%, TOV%, AST/G, BLK/G, STL/G, TOV/G

Context (3)

Playoff seed, late-season form (last 20 games), regular-season injury count

Target

Playoff_Rd: ordered 0-5 (missed → champion)

MODEL 1

Win % Alone

The simplest possible model. The baseline everything else must beat.

CROSS-VALIDATED R^2

0.581

Win percentage alone explains 58% of playoff variance.

Why It's a Strong Baseline

It encodes everything. Offense, defense, clutch performance, coaching, depth — all reduced to one number.

Standardized coefficient: **≈ 1.00** ($p < 0.001$)

A one-SD increase in win % (≈ 12-13 wins) predicts +1.0 playoff rounds.

The bar to clear: ***everything we add must beat 0.581.***

MODEL 2

+ 13 Advanced Stats

Net Rating, TS%, eFG%, defense, playmaking — does adding 13 features change everything?

Win % only

0.581

↓ *+ 13 features*

Cross-validated R^2

0.677

Gain: +9.6 percentage points

What Actually Mattered

Win % stays dominant (β = 1.00, p < 0.001).

Net Rating adds nothing beyond it (β = -0.09, p = 0.62); the two correlate at r = 0.97.
TS% is not significant (β = 0.16). 3P% turns slightly negative (β = -0.11, p = 0.013).

Steals, blocks, and assists show small, inconsistent effects.

Solid improvement. But not much better than last semester.

MODEL 3

+ Context Features

Playoff seed, late-season form, and regular-season injuries.

PLAYOFF SEED

SIGNIFICANT

β = -0.09 p = 0.004

Seeds run 1 to 8, with 0 for non-playoff teams, so the sign is hard to read directly.

LATE-SEASON FORM

WEAK

β = -0.03 p = 0.37

Small, and not statistically significant.

REGULAR-SEASON INJURIES

NEGLIGIBLE

β = 0.06 p = 0.045

Significant in-sample, but they add only 0.001 to cross-validated R^2.

CROSS-VALIDATED R^2

0.719

Gain: +4.2 percentage points. Context helps. Injuries add almost nothing.

DEEP DIVE

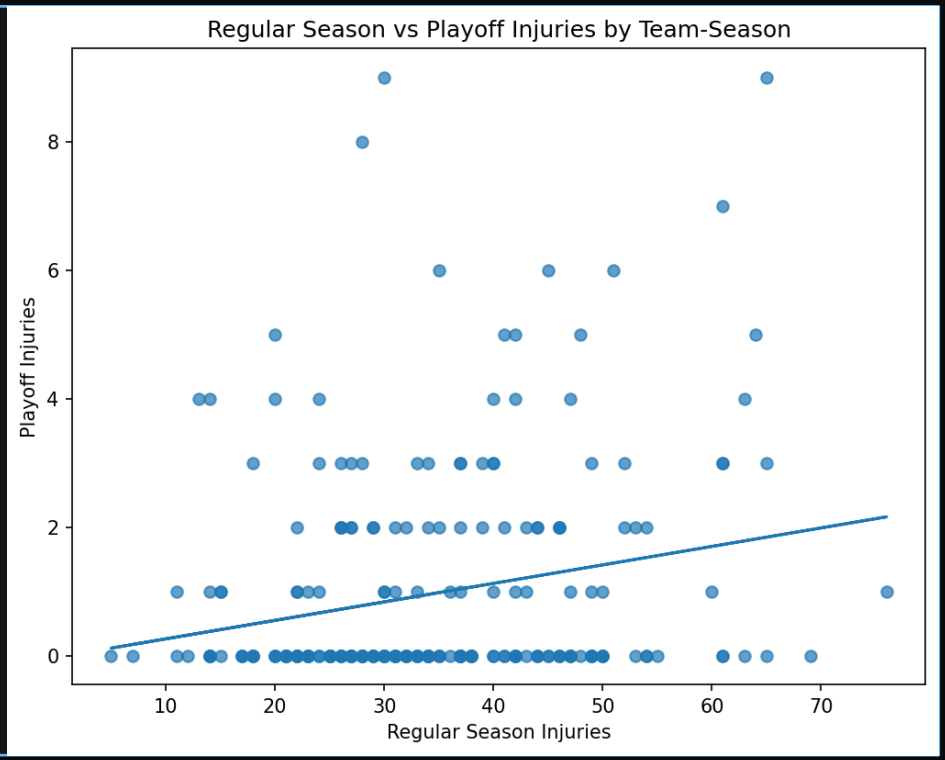
Why Did Injuries Add Almost Nothing?

Intuition says injuries should matter. The data says they don't. Here's why.

WHAT WE MEASURED VS. **WHAT ACTUALLY MATTERS**

Regular-Season Injuries

measure **durability under a different stress level**. A team that lost 10 games due to injury in December is not necessarily a team that loses Game 1 of the playoffs. Stars recover, role players come in and out. A healthy team can also lose its star in the first round of the playoffs. Unfortunately the two are not heavily correlated.



METHODOLOGY TEST

Does Machine Learning Help?

Same 17 features. Linear regression vs. XGBoost.

LINEAR REGRESSION

0.589

Assumes each feature contributes independently and additively. Misses interactions.

XGBOOST (TUNED)

0.720

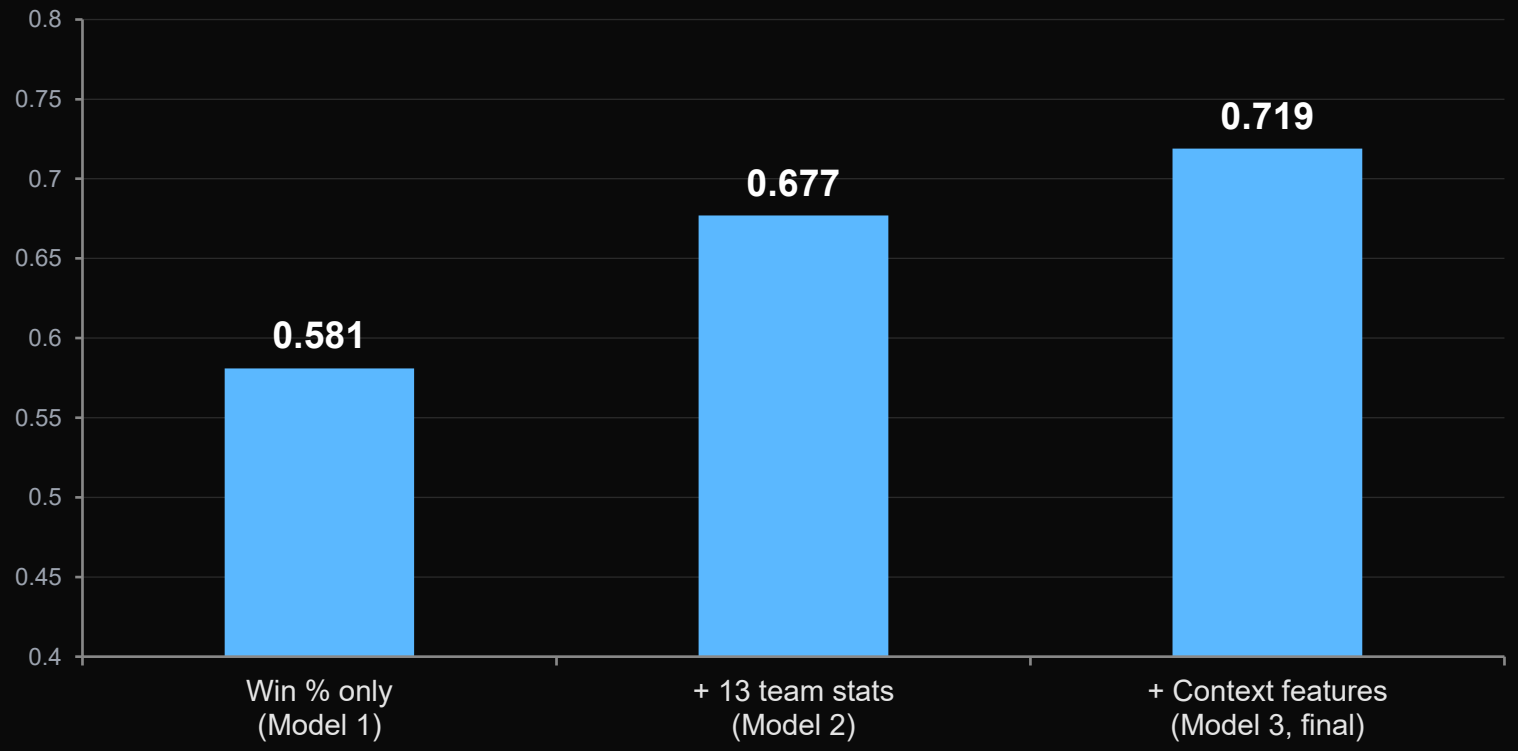
Captures non-linearities and feature interactions. Net Rating × seed. Form × elite tier.

Gain from ML: +13.1 points.

THE FINDING

Where the Gains Come From

Every step adds something. Engineered features add 13.8 points over win % alone.



DIMINISHING RETURNS

+9.6 win % → 14 features

+4.2 + context features

Final cross-validated $R^2 = 0.719$

Without ML, linear regression at 17 features only reaches 0.589. XGBoost found an extra +13.1 in non-linear interactions.

OUT-OF-SAMPLE TEST

2025 Holdout: Did the Thunder Win?

Trained on 1997-2024. Tested on a season the model had never seen.

0.559

HOLDOUT R^2

2nd-lowest of 29 seasons tested

53%

EXACT MATCHES

16 of 30 teams predicted correctly

87%

WITHIN ONE ROUND

26 of 30 teams within ±1 round

HITS

Oklahoma City Thunder → predicted to reach the Finals. They won the title.

Denver, Milwaukee, Memphis, Detroit → exact round predicted.

12 of 14 non-playoff teams correctly assigned zero.

MISSES

Cleveland Cavaliers → predicted champion, but they lost in the conference semifinals.

Indiana Pacers → predicted a round one exit, but they reached the finals; the classic hot playoff team.

What's in the Other 28%?

These factors are absent from regular-season data. No model can predict what isn't there.

Playoff Injuries

Regular-season durability ≠ playoff health. The 2019 Warriors lost Durant and Thompson in the Finals and went from favorites to losers.

Matchup Dynamics

Defensive schemes that exploit one offense may struggle against another. The regular season averages this; the playoffs magnify it across seven games.

Small-Sample Variance

Seven-game series are noisy. One hot shooting night, one bad call, one late-game mistake. Over 82 games, variance averages out. Over seven, it doesn't.

In-Series Adjustments

Coaches make mid-series strategic changes invisible in regular-season data. A team beaten by zone in February might solve it by Game 4.

THE TAKEAWAY

What can be improved?

With the data available, the model explains about 72% of playoff variance in cross-validation. What would happen if the model was updated daily?

Optimization

Models that use instantly updated stats can predict the outcome of games, even as they're happening. That's how Vegas makes their odds. A model with perfect information (all stats and news on the internet, new and old), should be able to predict not only the outcome of games, but beat Vegas more often than not. That is every analysts dream, and what this groundwork will grow into.

The model isn't perfect yet. ***But it's honest.***

Thank You

Questions?

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